GR Carbons — Dashboard KPI calculations

2026-10-06 · current live logic on the Activated Carbon dashboard. Formulas below are what the API and UI compute today, before any next-step change.

1. KPI table

KPI on screen	Meaning	Formula	Data used	Notes
Charcoal stock (Charcoal data)	Raw charcoal still at the dock — received, not yet released to production.	pending_kg = inward our-weight (ordered_kg) – allocated_to_production_kg	Inward orders (ORD-AC-*) with raw-material lines. KPI uses pending_kg. Chart uses each inward remaining available kg, stacked by supplier and material.	Does not include released lots. Chart and KPI can differ slightly if available kg and pending kg are not in sync.
GC (GCharcoal stock)	Released charcoal waiting to be loaded on a kiln.	Sum of lot_quantity for lots with status CREATED	Production lots created at Release to Production. Chart groups those lots by supplier and kiln.	GC is GCharcoal, not a carbon-number grade. Once the lot is loaded, it leaves GC and moves to Exklin with stone.
Plant yield	Plant-wide conversion of raw charged to finished output.	finished output kg ÷ raw charged kg × 100	Numerator: STOCKED lot kg + DISPATCHED last_dispatch_kg. Fallback: In Stock / finished-goods kg + all dispatch kg. Denominator: allocated_to_production_kg. Fallback: last_release_kg on the production order, then planned order qty.	Target on the gauge is 22%. Stays blank while material is only WIP (nothing stocked or dispatched). One plant number — not per lot on the card.
Production availability	Share of the active book that is currently running.	max(lots IN_PROGRESS, kilns RUNNING) ÷ active lots × 100	Active lots = lots not DISPATCHED / CLOSED / CANCELLED (or the dashboard active-order list). A kiln is RUNNING when an IN_PROGRESS lot is on it (IN_KILN, kiln feed, or activation).	Not equipment uptime or OEE. If there are no active lots, it uses kilns running ÷ kiln count. Target is 90%.
Material Flow (WIP)	Total kilograms sitting in the five stock stages.	Charcoal + GC + Exklin with stone + Exklin without stone + Final grade	Same five buckets shown on the Material Flow drill-down.	A kg snapshot, not a flow rate. Final grade is destoning / post-activation In Stock bags by grade, then stocked lots, then product FG kg.
Exklin with stone	Material on the kiln / shop floor — still in process.	Sum of lot_quantity where status is IN_PROGRESS, or pipeline stage is kiln feed, routing / execution, or execution (n/m) not complete	Lot status and pipeline_stage from the dashboard lot lines.	Name is a stage label. It is not destoning bag kg and does not add stones from crusher / destoning.
Exklin without stone	Shop-floor complete, not yet in finished-goods stock.	Sum of lot_quantity where status is COMPLETED or QC_PENDING, or stage is execution complete / shop-floor complete	Lot status and pipeline_stage from the dashboard lot lines.	Name is a stage label. It is not destoned (stone-removed) bag kg.
Kiln yield	Yield chart titled Kiln Yield (All Kilns).	Reuses the same plant yield %. Does not compute kiln A vs kiln B from feed vs output.	If two or more lots are STOCKED / DISPATCHED / QC_PENDING / COMPLETED, that one plant % is drawn on each of the last eight completed lots. Otherwise the same % is shown as a bar on every kiln.	Not a true per-kiln yield. Completing kiln output bags alone does not change this number until stock / dispatch exists.

2. How a lot moves across the stock stages

Exklin with / without stone follow lot status, not destoning capture. Pipeline stage is derived from lot status plus completed shop-floor operations.

Lot status	Pipeline stage shown	Dashboard bucket
CREATED	Awaiting kiln feed	GC (GCharcoal)
IN_PROGRESS	Kiln feed / Routing / Execution (n/m)	Exklin with stone
COMPLETED	Shop-floor complete	Exklin without stone
QC_PENDING	Quality testing	Exklin without stone
STOCKED	Finished goods stock	Final grade (if API grade stock is empty)
DISPATCHED	Dispatched	Not in Material Flow; used in plant yield numerator
CLOSED / CANCELLED	Closed	Ignored for active lots and Material Flow

3. Related numbers on the same dashboard

Related number	How it is calculated
Final grade stock	In Stock bag remaining kg by grade (destoning / post-activation destination In Stock). If none, stocked lot kg. If none, product finished-goods kg.
Kiln feed mass (drill-down)	GC kg + Exklin with stone kg.
Process loss	100 – plant yield %. Same plant yield figure, inverted.
Material balance	(charged_kg + pending_kg) ÷ ordered_kg × 100.
Kiln utilization	Kilns marked RUNNING ÷ kiln count × 100.